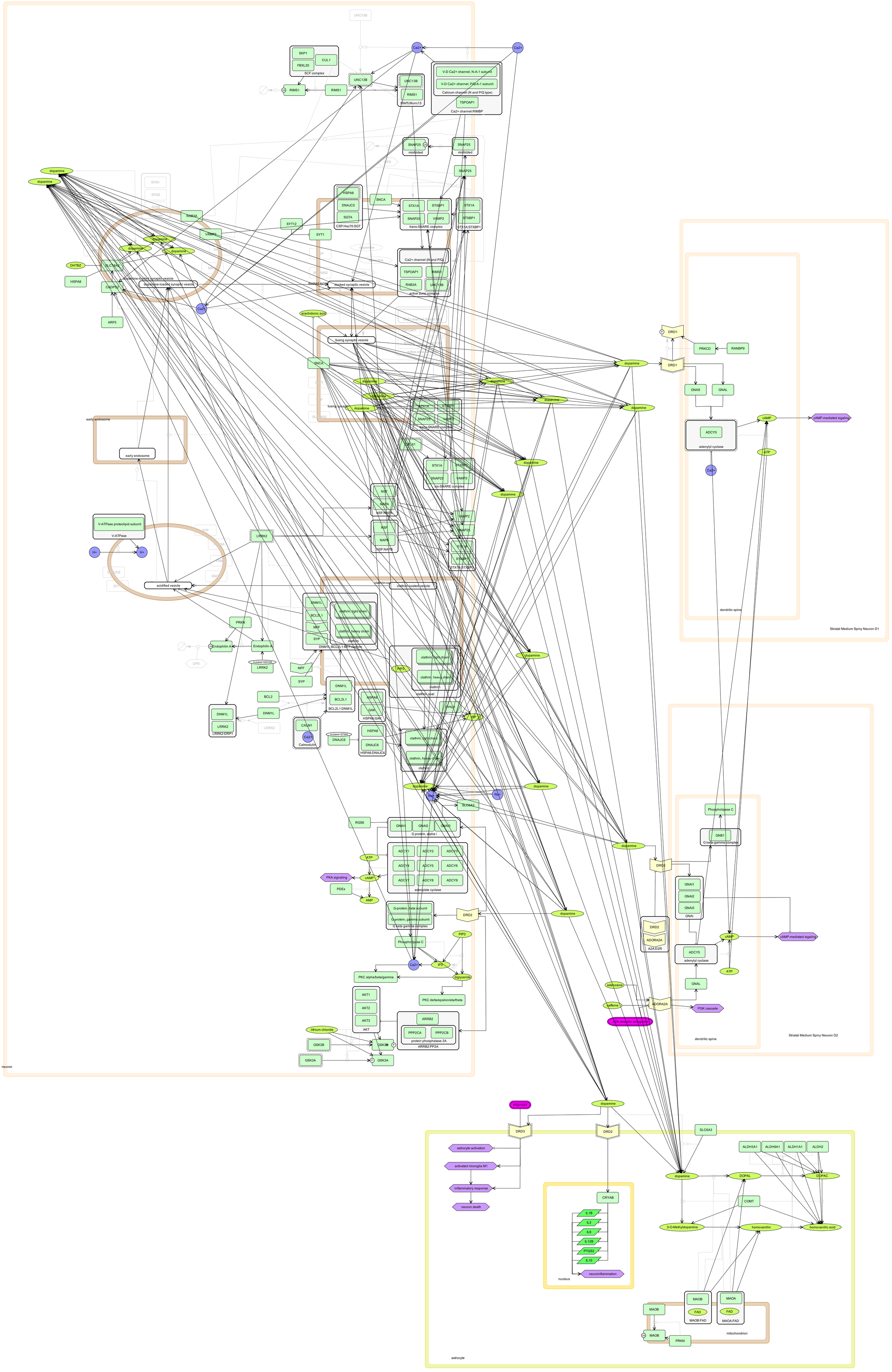

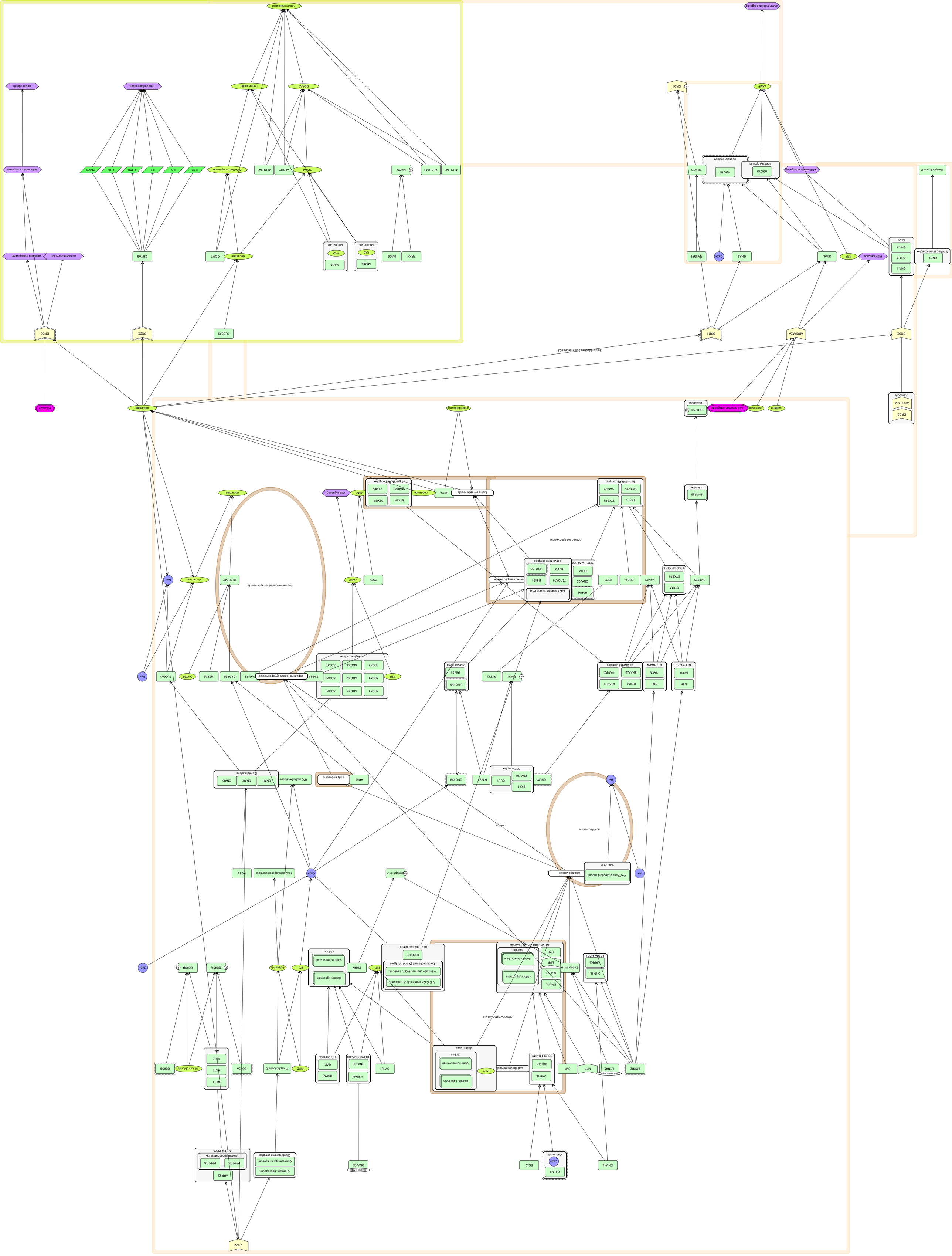

UNC13B
SKP1
CUL1
FBXL20
SCF complex
Ca2+
Ca2+
RIMS1
RIMS1
UNC13B
UNC13B
RIMS1
RIMS:Munc13
V-D Ca2+ channel, N-A-1 subunit
V-D Ca2+ channel, P/Q-A-1 subunit
Calcium channel (N and P/Q type)
TSPOAP1
Ca2+ channel:RIMBP
SNAP25
misfolded
SNAP25
misfolded
SNAP25
UPS
SYN1
SYN2
SYN3
dopamine
dopamine
HSPA8
DNAJC5
SGTA
CSP:Hsc70:SGT
SNCA
STX1A
STXBP1
SNAP25
VAMP2
trans-SNARE complex
STX1A
STXBP1
STX1A:STXBP1
RAB3A
VAMP2
SYT12
SYT1
dopamine
dopamine
dopamine
DHTBZ
SLC18A2
HSPA8
CADPS2
ARF5
dopamine-loaded synaptic vesicle
dopamine-loaded synaptic vesicle
docked synaptic vesicle
docked synaptic vesicle
Ca2+ channel (N and P/Q)
TSPOAP1
RIMS1
RAB3A
UNC13B
active zone complex
Ca2+
arachidonic acid
fusing synaptic vesicle
SNCA
dopamine
dopamine
dopamine
fusing synaptic
STX1A
STXBP1
SNAP25
VAMP2
trans-SNARE complex
CPLX1
early endosome
early endosome
STX1A
STXBP1
SNAP25
VAMP2
cis-SNARE complex
NSF
NAPA
NSF:NAPA
NSF
NAPB
NSF:NAPB
VAMP2
SNAP25
STX1A
STXBP1
STX1A:STXBP1
V-ATPase proteolipid subunit
V-ATPase
H+
H+
LRRK2
acidified vesicle
clathrin-coated vesicle
DNM1L
BCL2L1
MFF
SYP
clathrin, light chain
clathrin, heavy chain
clathrin
DNM1L:BCL2L1:MFF:clathrin
PRKN
Endophilin A
Ub
Endophilin A
UPS
mutated G2019S
LRRK2
MFF
SYP
PIP2
clathrin, light chain
clathrin, heavy chain
clathrin
clathrin coat
BCL2
DNM1L
DNM1L
BCL2L1
BCL2L1:DNM1L
HSPA8
GAK
HSPA8:GAK
SYNJ1
PIP
DNM1L
LRRK2
LRRK2:DRP1
LRRK2
CALM1
Ca2+
Calmodulin
mutated Q734X
DNAJC6
HSPA8
DNAJC6
HSPA8:DNAJC6
clathrin, light chain
clathrin, heavy chain
clathrin
dopamine
Na+
Na+
dopamine
SLC6A3
RGS6
GNAI1
GNAI2
GNAI3
G protein, alpha i
dopamine
ATP
ADCY1
ADCY2
ADCY3
ADCY4
ADCY5
ADCY6
ADCY7
ADCY8
ADCY9
adenylate cyclase
PKA signaling
cAMP
PDEs
AMP
G-protein, beta subunit
G-protein, gamma subunit
G beta gamma complex
DRD2
dopamine
PIP2
Phospholipase C
Ca2+
IP3
diglyceride
PKC alpha/beta/gamma
PKC delta/epsilon/eta/theta
AKT1
AKT2
AKT3
AKT
ARRB2
PPP2CA
PPP2CB
protein phosphatase 2A
ARRB2:PP2A
lithium chloride
GSK3B
GSK3B
P
GSK3A
P
GSK3A
neuron
DRD1
P
PRKCD
RANBP9
dopamine
DRD1
GNAS
GNAL
ADCY5
adenylyl cyclase
cAMP
cAMP-mediated sigaling
ATP
Ca2+
dendritic spine
Striatal Medium Spiny Neuron D1
Phospholipase C
GNB1
G beta-gamma complex
DRD2
GNAI1
GNAI2
GNAI3
GNAI
DRD2
ADORA2A
A2A:D2R
cAMP
cAMP-mediated sigaling
ADCY5
adenylyl cyclase
ATP
GNAL
adenosine
caffeine
ADORA2A
PI3K cascade
A2A receptor antagonist
dendritic spine
Striatal Medium Spiny Neuron D2
PG01037
dopamine
DRD3
DRD2
SLC6A3
astrocyte activation
activated microglia M1
inflammatory response
neuron death
CRYAB
IL1B
IL2
IL6
IL12B
PTGS2
IL10
neuroinflammation
nucleus
ALDH3A1
ALDH9A1
ALDH1A1
ALDH2
dopamine
DOPAL
DOPAC
COMT
3-O-Methyldopamine
homovanillin
homovanillic acid
MAOB
FAD
MAOB:FAD
MAOA
FAD
MAOA:FAD
MAOB
Ub
MAOB
PRKN
mitochondrion
astrocyte

DRD2
DNM1L
CALM1
Ca2+
Calmodulin
BCL2
mutated Q734X
DNAJC6
G-protein, beta subunit
G-protein, gamma subunit
G beta gamma complex
ARRB2
PPP2CA
PPP2CB
protein phosphatase 2A
ARRB2:PP2A
LRRK2
mutated G2019S
LRRK2
MFF
SYP
DNM1L
BCL2L1
BCL2L1:DNM1L
clathrin-coated vesicle
PIP2
clathrin, light chain
clathrin, heavy chain
clathrin
clathrin coat
SYNJ1
HSPA8
DNAJC6
HSPA8:DNAJC6
HSPA8
GAK
HSPA8:GAK
PIP2
Phospholipase C
GSK3A
AKT1
AKT2
AKT3
AKT
lithium chloride
GSK3B
clathrin-coated vesicle
DNM1L
LRRK2
LRRK2:DRP1
Endophilin A
DNM1L
BCL2L1
MFF
SYP
clathrin, light chain
clathrin, heavy chain
clathrin
DNM1L:BCL2L1:MFF:clathrin
V-D Ca2+ channel, N-A-1 subunit
V-D Ca2+ channel, P/Q-A-1 subunit
Calcium channel (N and P/Q type)
TSPOAP1
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PRKN
clathrin, light chain
clathrin, heavy chain
clathrin
IP3
diglyceride
P
GSK3A
GSK3B
P
Ca2+
H+
V-ATPase proteolipid subunit
V-ATPase
acidified vesicle
Ub
Endophilin A
Ca2+
PKC delta/epsilon/eta/theta
RGS6
acidified vesicle
neuron
H+
CPLX1
SKP1
FBXL20
CUL1
SCF complex
RIMS1
UNC13B
ARF5
early endosome
PKC alpha/beta/gamma
GNAI1
GNAI2
GNAI3
G protein, alpha i
NSF
NAPB
NSF:NAPB
NSF
NAPA
NSF:NAPA
STX1A
STXBP1
SNAP25
VAMP2
cis-SNARE complex
Ub
RIMS1
SYT12
UNC13B
RIMS1
RIMS:Munc13
ADCY1
ADCY2
ADCY3
ADCY4
ADCY5
ADCY6
ADCY7
ADCY8
ADCY9
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dopamine-loaded synaptic vesicle
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HSPA8
DHTBZ
SLC6A3
Na+
SNAP25
STX1A
STXBP1
STX1A:STXBP1
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STXBP1
SNAP25
VAMP2
trans-SNARE complex
fusing synaptic vesicle
SNCA
dopamine
STX1A
STXBP1
SNAP25
VAMP2
trans-SNARE complex
AMP
PKA signaling
dopamine
DRD2
ADORA2A
A2A:D2R
caffeine
adenosine
A2A receptor antagonist
SNAP25
Ub
misfolded
arachidonic acid
dopamine
PG01037
Striatal Medium Spiny Neuron D2
DRD2
ADORA2A
DRD1
SLC6A3
DRD2
DRD3
GNB1
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GNAI1
GNAI2
GNAI3
GNAI
PI3K cascade
ATP
GNAL
GNAS
Ca2+
RANBP9
PRKN
MAOB
MAOB
FAD
MAOB:FAD
MAOA
FAD
MAOA:FAD
dopamine
COMT
CRYAB
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activated microglia M1
Phospholipase C
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MAOB
DOPAL
ALDH2
ALDH3A1
3-O-Methyldopamine
IL1B
IL6
IL2
IL12B
IL10
PTGS2
inflammatory response
cAMP
P
DRD1
DOPAC
homovanillin
neuroinflammation
neuron death
cAMP-mediated sigaling
homovanillic acid